

Formalization of Thirty Three Miniatures

Notes

foo

foobar

1 Isabelle Questions

- Why does Abs_TYP just behave sensibly outside of its domain?
- What is a “qualified” lemma?

2 Isabelle Peculiarities

- Algebra structures are defined twice (once as type class, once as locale)

3 Math Stuff

- useful to realize implicit assumptions such as commutativity in a definition of sums

4 TODO: Potentially Interesting Talk Content

- Partially done: Overview over LA/Algebra in Isabelle (mention the chaos of everything being defined twice)
- Not done at all: Geometry reasoning in Isabelle
- Not done at all: Automation of LA proof methods
- Not done at all: Understand existing proof methods (argo is used a lot...?)
- Not done at all: ML for ?thesis in subgoals

5 Talk Structure?

1. Formalization content (general topic, interesting theorems)
2. Formalization tools (Isabelle/HOL, AFP theories)
3. Formalization struggles/observations
 - Content: necessary care for detail, easier to see potential for generalization/find errors and lack of detail in proofs/...
 - Isabelle: (Algebra) structures defined in multiple ways in different theories, reproving the same claims in the same proof twice?, excessive unfolding, no schematic variables for subgoals, ...
4. Isabelle adventures:
 - Eisbach proof methods
 - Isabelle/ML?